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DEPARTMENT OF GEOMATICS ENGINEERING

Banepa-05, Kavreplanchok

PROJECT REPORT ON



IMPACTS ON CORS OF NEPAL DUE TO EARTHQUAKE 2015

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ABSTRACT

The 2015 earthquake in Nepal had a significant impact on CORS (Continuously Operating Reference Station) in the country. The earthquake, which occurred on April 25, 2015, caused damage to several of the CORS, resulting in interruptions to data collection and transmission. The extent of the damage varied, with some stations being completely destroyed, while others were partially damaged. The impact of the earthquake on the CORS affected several sectors, including surveying, mapping, and engineering. The restoration and re-establishment of the affected CORS was a significant challenge due to the difficult terrain and limited resources in the region. However, efforts were made to repair the damaged stations, and the data collection and transmission have gradually resumed over time.

LIST OF ABBREVIATIONS

ASCII	American Standard Code for Information Interchange
ASPUS	Australian Positioning Reference System
BM	Benchmark
CIGNET	Cooperative International GPS Network
CIRES	Cooperative Institute for Research in Environmental Sciences
CMONOC	Crustal Movement Observation Network of China
CORS	Continuously Operating Reference Station
CSV	Comma-Separated Values
GAD	Geocentric Datum of Australia
GNSS	Global Navigation Satellite System
ITRF	International Terrestrial Reference Frame
NASA	National Aeronautics and Space Administration
NGS	National Geodetic Service
NOAAS	National Oceanic and Atmospheric Administration
NSF	National Science Foundation
NSRS	National Spatial Reference System
RS	Richter scale
RINEX	Receiver Independent Exchange Format
RTK	Real-time Kinematic
UNAVCO	University NAVSTAR Consortium

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1. INTRODUCTION

1.1 Background

A geodetic datum is an abstract coordinate system with a reference surface (such as Mean sea level) that serves to provide known locations to begin surveys and create maps where a horizontal datum is used to measure a location across the Earth's surface, in latitude (ϕ) and longitude (λ) and a vertical datum (h) is used to measure the elevation or depth relative to a standard origin. Modern geodetic datum is used for plane surveying to complex models used for international applications which completely describe the size, shape, orientation, gravity field, and angular velocity of the earth. The datum is shift due to various reasons such as sudden movement due to Earthquake, Volcanos.

The devastating Gorkha Earthquake measuring to 7.8 magnitude having the epicenter occurred near Barpak village of Gorkha District which is 181 KM northwest of Kathmandu on 25th April 2015 which kills around 9,000 people, of whom approximately 55 percent were female, and over 22,000 people were injured. The Earthquake occurred on the Main Himalayan Thrust Fault. This happened due to when two blocks of Earth suddenly slipped past one another. The surface where they slip is called fault or fault plane. The 2015 Gorkha Nepal earthquake caused tremendous damage and loss in lives and physical infrastructure along with the huge loss in historical, Social, cultural and economic aspects of Nepal. Among 77 districts, 14 of them are highly affected. This indicates that sufficient structural design is the key to reduce the earthquake risk in Nepal. The earthquake shifted the control points of Nepal such as in direction, elevation. The movement of Control points is the steady and slow motion of a portion of the earth area with relation to each other due to the movement of plate underneath. The abrupt motion of the control point along with ground by earthquake is called the movement of control points in plan metric and elevation.

Control point is an accurately surveyed coordinate location for a physical feature that can be identified on the ground. Control points are used in least-squares adjustments as the basis for improving the spatial accuracy of all other points to which they are connected. The movement of Control points is the steady and slow motion of a portion of the earth area with relation to each other due to the movement of plate underneath. The abrupt motion of the control point along with ground by earthquake is called the movement of control points in plan metric and elevation.

CORS is a network of stations that provides continuous GNSS data consisting of carrier and pseudo range measurements. The CORS data can be used for processing the shifted coordinates caused by seismic activities. The analysis of CORS coordinates is carried out for various epochs of time which provides comparison between coordinates prior and later the earthquake. This comparison provides insights on magnitude and direction of shifts on CORS.

The UNAVCO stand for University NAVSTAR Consortium. UNAVCO is a non-profit university-governed consortium that facilitates geoscience research and education using Geodesy.

AUSPOS is an acronym that may refer to the Australian Positioning Reference System. It is a GNSS (Global Navigation Satellite System) network of permanent GNSS stations in Australia

that provide real-time and post-processed GNSS data for various applications, including surveying, geodesy, and mapping.

1.2 Problem Statement

After the 2015 earthquake, different research and study finds the changes in different sectors so this project helps to find the shift of the control points of Nepal. The control points of Nepal established by the Survey Department of Nepal have not been revised and updated continuously. The slow crustal movement and other external forces cause shifts in the established control points.

1.3 Objectives

The Primary objective of the project is to study the impact and shift of control points by the Gorkha earthquake 2015.

The Secondary objectives of the project are as follows:

- To determine northing, easting and vertical shifts of each CORS station.
- To analyze the data downloaded from UNAVCO. Represents the Analyzed data of UNANCO.
- To ensure that the CORS network is fully functional and capable of supporting ongoing and future surveying projects in Nepal, including infrastructure development and disaster risk management.

1.4 Scope

This project covers the areas that were affected by the Gorkha Earthquake 2015.

The impacts of the 2015 Nepal earthquake on CORS networks were significant and have had lasting effects on the accuracy and reliability of geodetic measurements in the region. However, efforts are underway to repair and improve the CORS network, including the installation of new stations in areas where there were previously none.

2. Literature Review

2.1 Earthquake

Earthquake means any sudden shaking of the ground caused by the passage of seismic waves through Earth's rocks. Seismic waves are produced when some form of energy stored in Earth's crust is suddenly released, usually when masses of rock straining against one another suddenly fracture and "slip". Earthquakes occur most often along geologic faults, narrow zones where rock masses move in relation to one another. The major fault lines of the world are located at the fringes of the huge tectonic plates that make up earth crust. The magnitude of the earthquake is measured in the Richter Scale(RS).

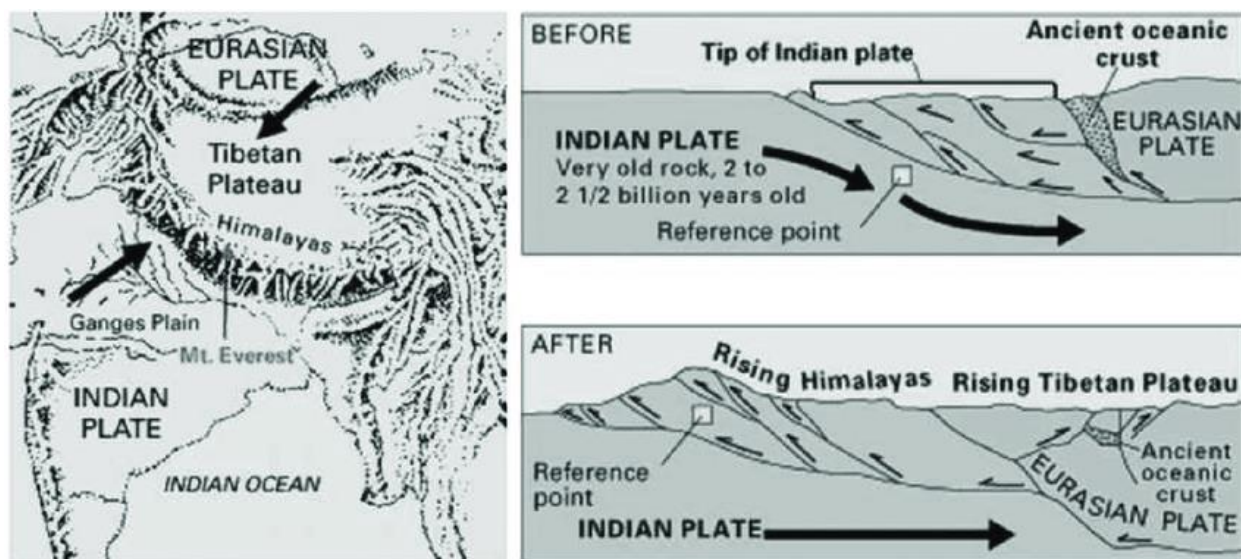


Figure 1: Interaction between Indian and Eurasian plates

2.2 Historical Background of Earthquake

Nepal is the 11th most Earthquake-prone Country in the World. The first recorded Earthquake in the history of Nepal took place on June 7, 1255 A.D. The magnitude of the Earthquake is said to be around 7.7 in magnitude. One third of the total population of Kathmandu were killed including Abahya Malla. In 1408 A.D, 1681 A.D and 1810 A.D massive earthquakes struck Nepal. In 1833 A.D another Earthquake devastated Kathmandu valley. Its magnitude was reported to be 7.8. The event was located 50 Km North-North East of Kathmandu. In 1934 A.D 8 magnitude Earthquake occurred in the territory of Nepal with casualties of more than 16000 people. In 1988 A.D Udayapur Earthquake. On 25th April, 2015 at 11:56 am 7.8 magnitude Earthquake occurred near Barpak village of Gorkha District which is 80 Km North-North East of Kathmandu and 6.7 magnitude Earthquake occurred on the same day. On April 26, 2015, 6.9 magnitude and on May 12, 2015, 7.3 magnitude of Earthquake was recorded.

2.3 Socio-Economic Effects of the Gorkha Earthquake 2015

The Gorkha Earthquake of April 25, 2015 caused huge effect on lives, infrastructure and livelihood of people. Out of 77 districts, 31 districts are affected and the damage of 14 districts are listed below: -

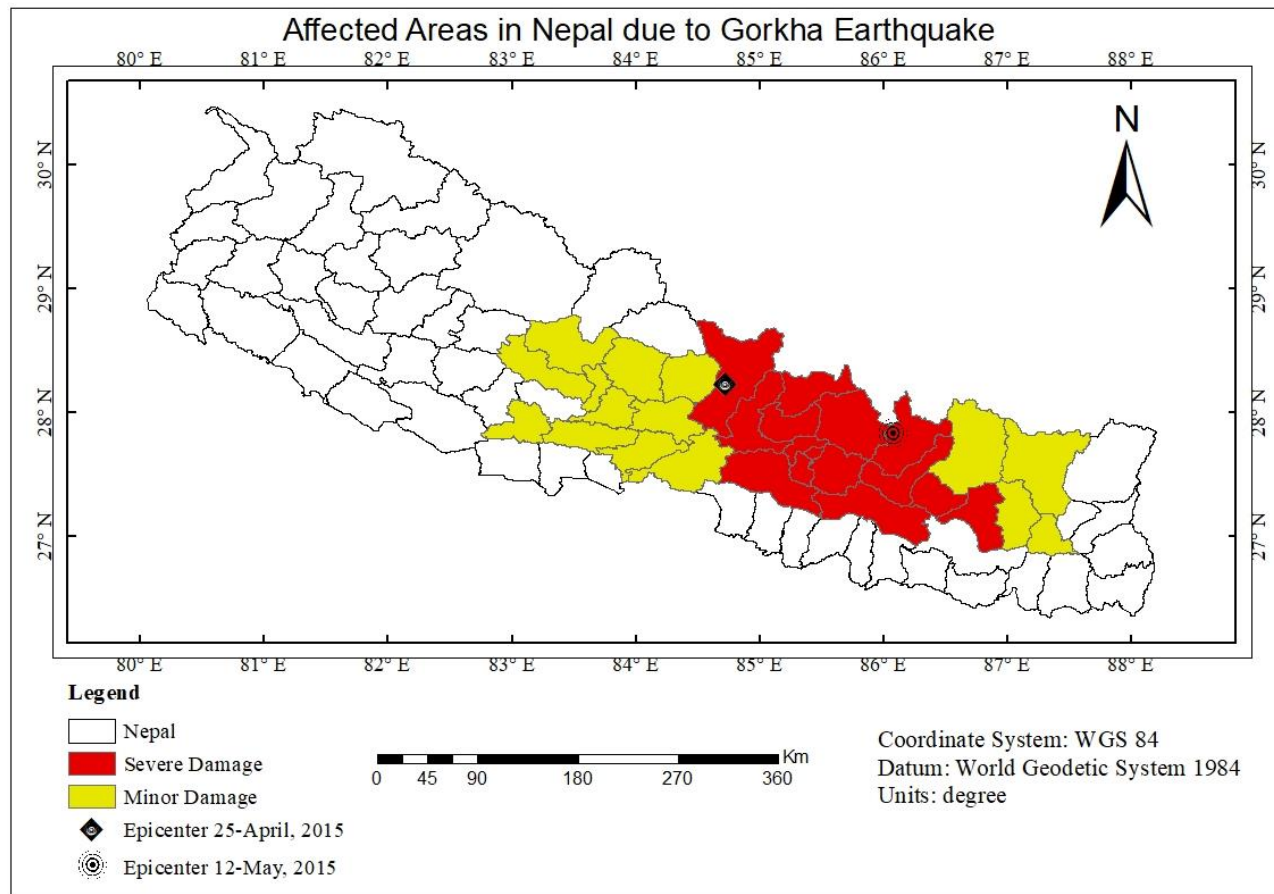


Figure 2: Districts of Nepal effected by earthquake of 2015

The table below shows a detail of damage caused by Gorkha Earthquake 2015 and aftershocks on households, lives and infrastructures.

Table 1: Details of Damage in Districts of Nepal

S.N.	District	Fatalities		Injuries	Number of Residential Building	
		Male	Female		Collapsed	Partially Damage
1	Sindhupalchwok	1497	1943	1753	63885	2751
2	Kathmandu	622	600	7952	43502	56024
3	Lalitpur	69	108	3051	17444	8064
4	Bhaktapur	119	214	2101	18900	9054
5	Gorkha	215	233	952	59527	13428
6	Sindhuli	5	10	230	18197	10028
7	Rasuwa	312	344	771	11368	267
8	Okhaldhunga	10	10	61	10031	3107
9	Makwanpur	16	17	229	20035	17383
10	Kavrepalanchok	129	189	1179	49933	223714
11	Dhading	340	393	1218	81313	3092
12	Nawakot	461	638	1050	75562	4200
13	Dolakha	84	85	662	48880	3120
14	Ramechhap	17	23	134	267423	13173

2.4 Control Point

An accurately surveyed coordinate location for a physical feature that can be identified on the ground. Controls are used in least squares adjustment as the basis for improving the spatial accuracy of all other points to which they are connected. Plan metric" and "altimetry" are terms used to describe the measurement of features on a map or in a geospatial dataset. They are listed below: -

- Plain Metric
- Altimetry

2.4.1 Plain Metric

Plan metric details refer to the measurement of features in a two-dimensional plane, such as the position of a building or a road on a map. These details provide information about the horizontal location and shape of features but not their vertical position or elevation.

2.4.2 Altimetry

Altimetry details, on the other hand, refer to the measurement of the elevation or height of features, such as mountains or valleys. These details provide information about the vertical position of features and are usually depicted on a map using contour lines or shading to indicate changes in elevation.

2.5 University NAVSTAR Consortium (UNAVCO)

The University NAVSTAR Consortium (UNAVCO) is a non-profit, university-governed consortium that supports research and education in the geosciences. UNAVCO operates a network of geodetic instruments, including Global Positioning System (GPS) and other geodetic technologies, that provide high-precision measurements of the Earth's surface and its motion. These measurements are used by researchers to study the Earth's crustal deformation, plate tectonics, and other geologic processes.

UNAVCO was established in 1984 by a group of geoscientists from universities in the United States who recognized the need for a network of geodetic instruments to support their research. Today, UNAVCO is funded by the National Science Foundation (NSF) and includes more than 100 member institutions, including universities and research organizations from around the world.

In addition to operating the geodetic network, UNAVCO provides training and education programs for researchers and students, supports scientific field expeditions, and collaborates with other organizations to advance the field of geodesy. UNAVCO's data and services are freely available to the scientific community and the general public, promoting collaboration and scientific progress in the geosciences.

Website: www.unavco.org

2.6 GNSS

UNAVCO operates a network of geodetic instruments, including Global Navigation Satellite System (GNSS) receivers, that provide high-precision measurements of the Earth's surface and its motion. UNAVCO's GNSS network includes more than 1,300 continuously operating receivers located throughout North America, South America, and the Arctic, as well as additional receivers deployed for specific research projects around the world.

The GNSS data collected by UNAVCO's network is used by researchers to study plate tectonics, earthquake hazards, crustal deformation, and other geologic processes. The data is also used to support applications such as precise positioning for surveying and mapping, and to improve our understanding of the Earth's atmosphere and climate.

UNAVCO provides a range of GNSS services and support, including data access, data processing and analysis, and technical support for instrument installation and operation.

UNAVCO's GNSS data is freely available to the scientific community and the public through several data portals, including the UNAVCO Data Archive and the Global Navigation Satellite System (GNSS) Data and Products portal.

Overall, UNAVCO's GNSS network and services play a critical role in advancing the field of geodesy and supporting research in the geosciences.

2.7 COR Station

A Continuously Operating Reference Station(CORS) network is a network of RTK base stations that broadcast corrections, usually over an Internet connection. The CORS network is a multipurpose, multi-agency cooperation endeavor, combining the efforts of hundreds of government, academic and private organizations.

Continuously Operating Reference Stations (CORS) are permanent GNSS operating stations mounted on stable platforms across a wide area and meant to serve as reference or base stations for stand-alone rovers or receivers within the coverage of the transmitted signal from the CORS. A COR station is occupied by a GNSS receiver that collects GNSS data continuously. CORS network is available 24 hours per day, 7 days a week and 365 days a year. However, it depends on the communication Service Provider and satellite availability. The collected data is streamed to a central coordinating center. Surveyors, GIS users, engineers, scientist and the public at large that collects GNSS data can use CORS data to improve the precision of their position CORS is an economical method (a reduction in both personnel and equipment) developed to utilize the free, static GNSS of data from the network of CORS (11th FIG Symposium on Deformation Measurements, Santorini, Greece, 2003). The GNSS data (RINEX file format) from CORS are downloaded and archived daily and are available free of charge to users.

CORS observations have been essential to study the nature of the 2015 Gorkha earthquake and associated deformations. A total of 33 CORS stations observations that spanned before and after earthquake time point and confirmed no transient preseismic displacements but found clear post seismic displacements Huang et al. used CORS stations from the Nepal region and Crustal Movement Observation Network of China (CMONOC) observations to study rupture propagation of the Gorkha earthquake and coseismic displacements. This earthquake is the largest continental thrust earthquake that has been recorded by 5 Hz GPS observations.

In Nepal the active CORS were identified between the date April 18, 2015 and May 18, 2015 which covered the two major earthquakes occurred during that time. The list of all active CORS in this period were: They are listed below: -

Table 2: COR STATION in Nepal

S.N	CODE	Name of Location	Remarks
1	AIRP	Tribhuban International Airport	Active
2	BELT	Beltari	Active
3	BESI	Besisahar	Active
4	BNDP	Bandipur	Active
5	CHLM	Chilime	Active
6	CHWN	Chitwan	Active
7	DMAU	Damuli	Active
8	DNSG	Dansing	Active
9	GHER	Ghermu	Active
10	JMSM	Jomsom	Active
11	LMJG	Lamjung	Active
12	NAST	KATHMMANDU	Active
13	OKRI	OKRI	Active
14	RAMTE	Ramite	Active
15	SIM4	Simara	Active
16	SNDL	Sindhuli	Active
17	SYBC	Syangboche	Active
18	KKN4	Kakani 4	Active
19	KIRT	Kirtipur	Active



Figure 3: COR Station

2.8 Continuously Operating Reference Stations (CORS): Historical Development

The history of the Continuously Operating Reference Stations (CORS) system is intimately connected to the National Oceanic and Atmospheric Administration (NOAAS) National Geodetic Survey (NGS) and this agency's mission is to define, maintain and provide access to the U.S. National Spatial Reference System (NSRS). The NSRS contains all necessary information about orientation and scale relative to International Reference Frames. The National Geodetic Service (NGS) recognized the potential contributions of GNSS for enhancing the NSRS in the early stages of GNSS development. Hence in the late 1980s, these agencies embarked on applying instrumentation and field techniques to improve the NSRS.

NGS in the fall of 1986, introduced the Cooperative International GPS Network (CIGNET), the forerunner of the CORS network. Each CIGNET Site was equipped with high quality dual frequency GPS receiver that continuously recorded signals from satellites. The primary intention GNSS was to make dependable tracking data available from a network of ground stations to compute precise ephemerides (orbits) for the GPS satellites. In 1989, CIGNET three stations in

the United States (MOJA in Mojave, Calif; RICH in Richmond, Fla; and West in Westford, Mass). These early CORS were equipped with Mini-Mac 2816-At dual frequency codeless receivers. By the end of 1991, CIGNET comprised a total of 21 sites spanning all continents except Antarctica. NGS augmented the CIGNET network, generating the core of the first public global GPS network that, unknowingly at the time, evolved into the current International Global Navigation Satellite System (GNSS) service (IGS) under the auspices of the international Association of Geodesy. As of today, there are 1,350 Continuously Operating Reference Stations that are fully operational worldwide.

2.9 Receiver Independent Exchange Format (RINEX)

RINEX is an internationally recognized format that is developed by the GNSS community for observation data, navigation data, clock data and many other data types. RINEX is a data interchange format for raw satellite navigation system data. Hatanaka File. The Hatanaka compressed ASCII format version of RINEX observation file is frequently used in conjunction with the UNIX compress, zip, gzip or other generalized compression utilities to create a very small file for internet transformation.

2.10 Background of AUSPOS

AUSPOS was established in response to the growing demand for accurate and reliable positioning information in Australia, especially for geospatial applications such as surveying, mapping, and infrastructure development. The system was set up in the late 1990s by Geoscience Australia to provide a national network of GNSS reference stations that could be used to support the needs of the surveying and mapping community. Since its establishment, the AUSPOS network has expanded to over 200 GNSS reference stations across Australia, covering all states and territories, and providing coverage to remote and regional areas. The system is continuously monitored and maintained by Geoscience Australia to ensure high-quality GNSS data is available to users. The AUSPOS network provides real-time GNSS data as well as post-processed GNSS data, which can be used to determine accurate positions with high precision. This data is made available to the public through the AUSPOS website, where users can access GNSS data from the network of reference stations and use it to support their geospatial applications. Overall, the AUSPOS network has played a critical role in supporting the growth of the geospatial industry in Australia and has helped to ensure the continued accuracy and reliability of geospatial data in the country.

2.11 AUSPOS

AUSPOS is an acronym that may refer to the Australian Positioning Reference System. It is a GNSS (Global Navigation Satellite System) network of permanent GNSS stations in Australia that provide real-time and post-processed GNSS data for various applications, including surveying, geodesy, and mapping. The AUSPOS system is maintained by Geoscience Australia, the national geological survey agency of Australia. AUSPOS is a free online GPS data processing service provided by Geoscience. AUSPOS takes advantage of both The IGS Stations

Network and the IGS product range, and works with data collected anywhere on Earth. Users submit their dual frequency geodetic quality GPS RINEX data observed in a 'static' mode to the AUSPOS GPS data processing system. An AUSPOS report will be emailed to you (typically in less than ten minutes) containing Geocentric Datum of Australia 202 (GDA2020) 1994 (GDA94) and International Terrestrial Reference Frame (ITRF) coordinates.

Website: <https://www.ga.gov.au/scientific-topics/positioning-navigation/geodesy/auspos>

2.12 How does AUSPOS works?

A standalone receiver finds its position by relying on the raw data obtained from satellites only. In this case, it can determine its position at the several-meter level. To improve the accuracy, the receiver can be used with a base station in the RTK setup. Since both receivers work in the same conditions, the base station sends corrections that eliminate ionospheric and tropospheric delays, clock errors of the receiver and satellites and improve accuracy. By the help AUSPOS, we can achieve a centimeter-level accurate position of a point with one receiver. When users submit their dual frequency geodetic quality GPS RINEX observation file in a 'static' mode to the AUSPOS GPS data processing system, AUSPOS chooses up to 15 nearest International GPS Service (IGS) and Asia-Pacific Reference Frame (APREF) stations for processing. These stations form a local network that provides the data required to eliminate ionospheric and tropospheric delay errors. With the help of the data from these stations, the service calculates the approximate coordinates of the receiver. The last stage of the processing is computing a solution. AUSPOS eliminates clock errors of the receiver and the satellites to obtain precise coordinates of the point. An AUSPOS report will be emailed to you (typically in less than ten minutes) containing Geocentric Datum of Australia 202 (GDA2020) 1994 (GDA94) and International Terrestrial Reference Frame (ITRF) coordinates.

2.13 Hatanaka File

The Hatanaka compressed ASCII format version of RINEX observation file is frequently used in conjunction with the UNIX compress, zip, gzip or other generalized compression utilities to create a very small file for internet transformation.

3. METHODOLOGY

3.1 Study Area

The study area map of our project shows the location of CORS station of Nepal. Below map shows the position of the stations whose data were used in order to get our result. For this purpose, the ground based dual frequency CORS situated along the different parts of Nepal were considered as a data source.

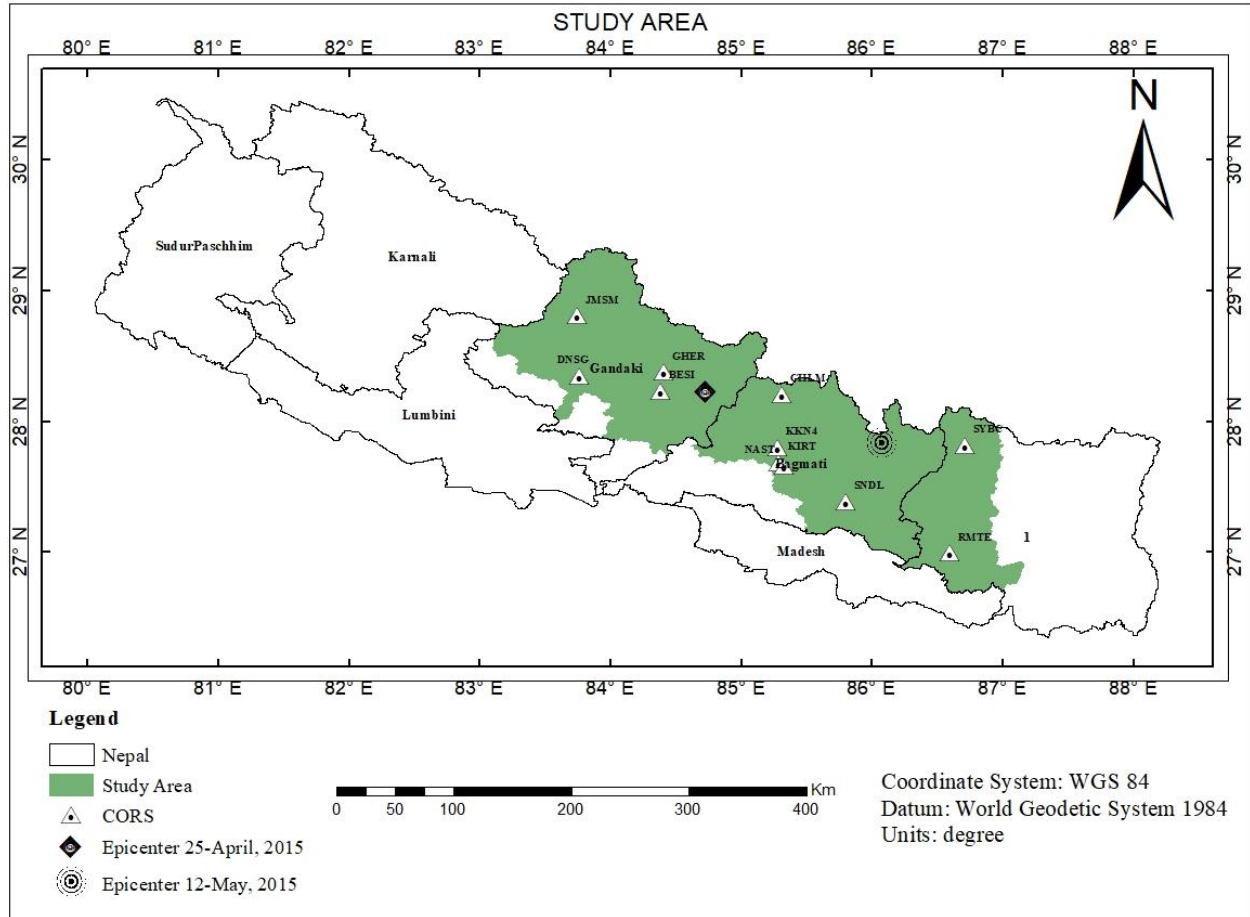


Figure 4: Study Area

3.2 List of Software

- **Arc GIS:** Arc Map was for the purpose of interpolation, visualization and map layout.
- **Microsoft office packages:** MS Excel was used for the data manipulation, analysis, management and calculation purpose and MS word for the purpose of documentation and reporting.

3.3 Methods and Workflow

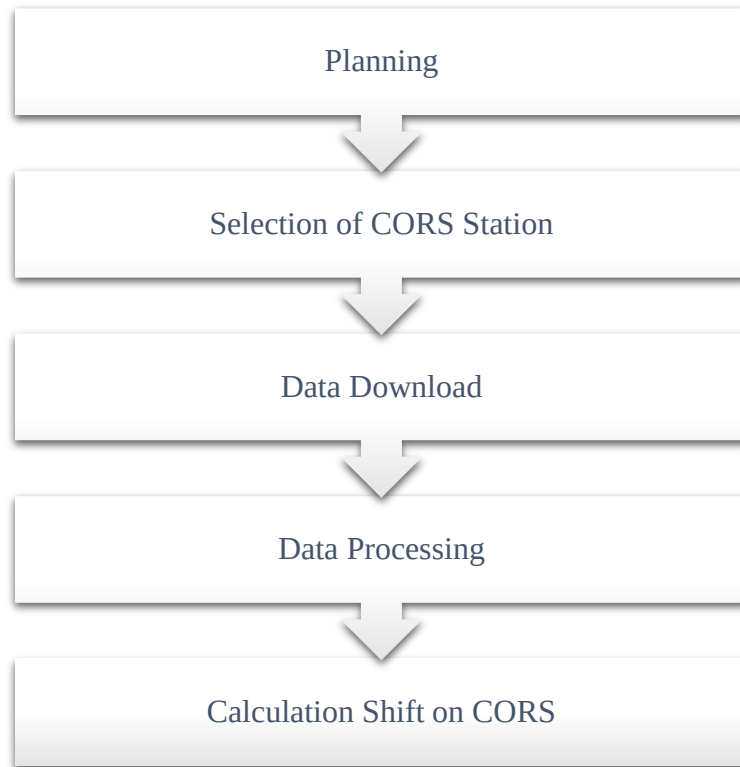


Figure 5: Methodology Flowchart

➤ **Planning:**

This is the first stage of our project. In the planning phase, the work to be done is discussed. It covers schedule, budget, time, etc. First of all, we plan to select a CORS and study it. Then, we move forward for detailed study of CORS data which are downloaded from UNAVCO. Then we make a project schedule and divide the work to perform. As well as, we plan for a budget and manage a time Schedule to finish tasks on an estimated time. We select the CORS before earthquake 2015/April/24 and after earthquake 2015/April/26.

➤ **Selection of CORS Station:**

All the active CORS were obtained during that period from UNAVCO website. From all the active CORS, we collect the list of COR Station. Then, the CORS are selected which is more affected by the Gorkha Earthquake 2015.

Table 3: List of COR station Used in the project

S.N.	Name of Station	Location
1	BESI	Besisahar
2	CHLM	Chilime
3	DNSG	Dansing
4	GHER	Ghermu
5	JMSM	Jomsom
6	KIRT	Kirtipur
7	KKN4	Kakani 4
8	NAST	Kathmandu
9	RMTE	Ramite
10	SNDL	Sindhuli
11	SYBC	Syangboche

➤ **Data Download:**

The raw data will be downloaded from the UNAVCO website as per the need. The hatanaka files will be downloaded for the processing. The downloaded files are zipped from which files will then be extracted. The website interface that was used to download data is <https://www.unavco.org/data/gps-gnss/data-access-methods/dai1/dai1>

➤ **DATA Processing:**

The Australian Positioning Reference System (APRS) is a geodetic framework that provides a standard reference system for positioning and mapping in Australia. It is based on the Geocentric Datum of Australia (GDA) and is used to convert data between different coordinate systems. To convert data using APRS, the first step is to determine the coordinate system of the data you want to convert. Then, you can use software or online tools that support APRS to perform the conversion.

For example, if you have latitude and longitude coordinates in the WGS84 datum (which is commonly used by GPS systems), you can use APRS to convert them to the GDA datum used in Australia. This conversion will take into account the differences in the shape and orientation of the earth's surface in different parts of the world, so that the resulting coordinates are accurate and consistent with other mapping data in Australia. Overall, the APRS provides a standardized framework for mapping and geodetic

surveying in Australia, and helps ensure that location data is accurate and consistent across different systems and applications.

The UNAVCO GPS/GNSS RINEX data file was processing undertaken by the AUSPOS Online GPS Processing Service. In this is the method of computed coordinates, ITRF2014 and there was some Cartesian ITRF2014 its represent in (X, Y, Z), Geodetic GRS80 Ellipsoid, ITRF2014 its represent in (Longitude, Latitude, Ellipsoid height) & UTM Grid, GRS80 Ellipsoid, ITRF2014 its represent in (Easting, Northing, Ellipsoid height) so we use the UTM Grid GRS80 Ellipsoid.

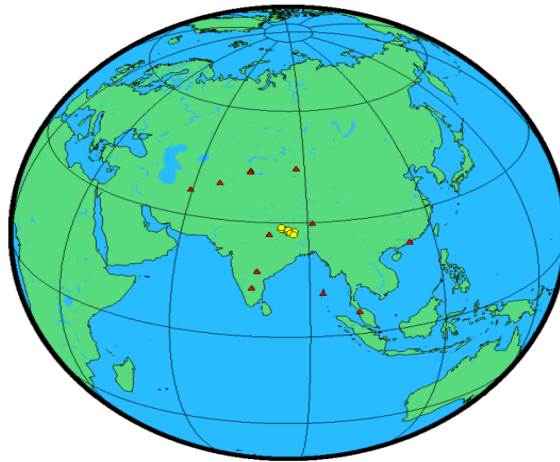


Figure 6: User Station and Reference Station

Table 4: User Station & Reference Station

Date	User Stations	Reference Station	Orbit Type
2015/04/24 2015/04/26	BESI, CHLM, DNSG, GHER, JMSM, KIRT, KKN4, NAST, RMTE, SNDL, SYBC	CHUM, DSMG, FOMO, HYDE, IISC, KIT3, LCK3, LHAZ, MSHN, PBRI, POL2, TOKA, URUM	IGS Final

➤ **Calculation Shifts on COR Station:**

The shift of the station will be computed after data is analyzed, obtained or converted. For the computation of shifts due to the earthquake, we will subtract the coordinates of every COR Station after the earthquake to before the earthquake 2015.

4. RESULTS

The results of project are:

- A. The processed coordinates for the stations recorded by CORS in 2015. The tabulated from of the coordinates in the WGS 1984 UTM Zone 45N projected coordinate system are:
- Before Earthquake, 2015/ April/24

Table 5: Coordinates of CORS before Earthquake, 2015/04/26

S.N	Station	Easting(m)	Northing(m)	Height (m)
1	BESI	242858.338	3125309.402	736.602
2	CHLM	334541.637	3121309.675	3211.621
3	DNSG	770898.373	3138533.853	2425.918
4	GHER	246156.253	3141428.012	1528.937
5	JMSM	767751.624	3189502.508	3436.805
6	KIRT	331181.971	3063135.098	1275.725
7	KKN4	330443.118	3076316.156	1895.739
8	NAST	335046.431	3060290.381	1276.7443
9	RMTE	460018.856	2985496.572	2072.064
10	SNDL	381131.722	3029632.376	2001.275
11	SYBC	471679.839	3076657.654	3792.766

➤ After Earthquake, 2015/April/26

Table 6: Coordinate of CORS after Earthquake, 2015/04/26

S.N	Station	Easting(m)	Northing(m)	Height (m)
1	BESI	242858.334	3125309.379	736.584
2	CHLM	334541.397	3121308.29	3211.032
3	DNSG	770898.372	3138533.849	2425.909
4	GHER	246156.274	3141427.963	1528.944
5	JMSM	767751.629	3189502.501	3436.801
6	KIRT	331181.602	3063133.645	1276.459
7	KKN4	330442.649	3076314.332	1897.004
8	NAST	335046.099	3060289.093	1277.058
9	RMTE	460018.868	2985496.571	2072.069
10	SNDL	381131.768	3029631.151	2001.301
11	SYBC	471679.835	3076657.638	3792.758

B. The next result of the project is the calculated shift in the easting, northing and height component of each stations considered. The shift due to the massive earthquake of April 25, 2015 & total shift was calculated which is tabulated below:

Table 7: Coordinate of Shift of each station used in the project

S.N.	Stations	dE(m)	dN(m)	dH(m)
1	BESI	-0.004	-0.023	-0.018
2	CHLM	-0.24	-1.385	-0.589
3	DNSG	-0.001	-0.004	-0.009
4	GHER	0.021	-0.049	0.007
5	JMSM	0.005	-0.007	-0.004
6	KIRT	-0.369	-1.453	0.734
7	KKN4	-0.469	-1.824	1.265
8	NAST	-0.332	-1.288	0.3137
9	RMTE	0.012	-0.001	0.005
10	SNDL	0.046	-1.225	0.026
11	SYBC	-0.004	-0.016	-0.008

C. The eastern part of the epicenter (Barpak) was hugely affected by the earthquake whereas the western part had minor effect. The station DNSG had the minimum vertical shift of -0.001m and the station NAST had the maximum vertical shift of -1.824m and the station SNDL had the minimum horizontal shift of 0.046m and the station KKN4 had the maximum horizontal shift of -0.469m because of the Gorkha Earthquake 2015 and the aftershock. The graphs showing change in Easting and Northing of each stations are shown below:

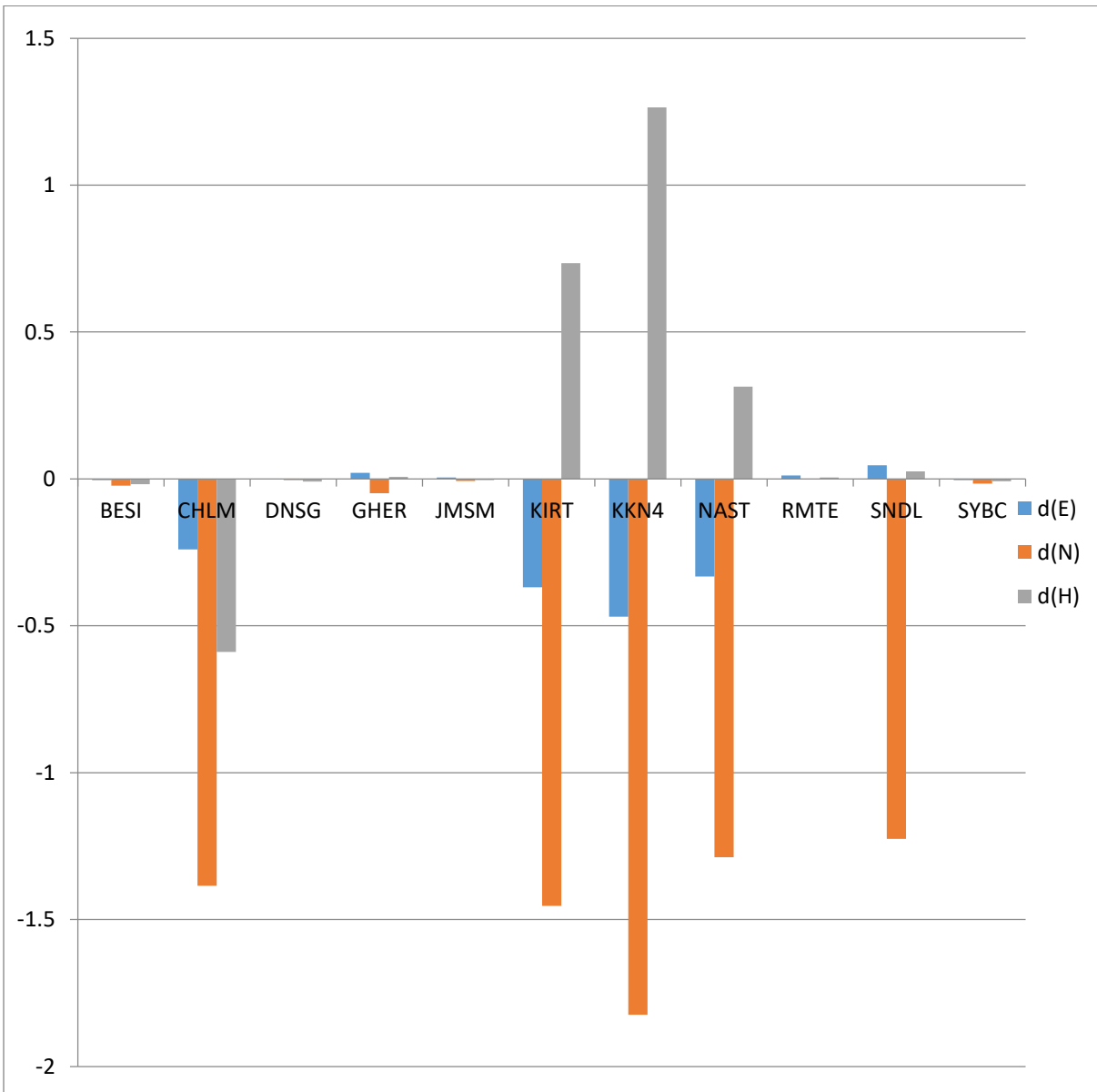


Figure 7: Graph showing Shift Direction of each Station

- D. The next result of the project is shift due to the massive earthquake of April 25, 2015 & total shift was map visualization as shown in below the figure.

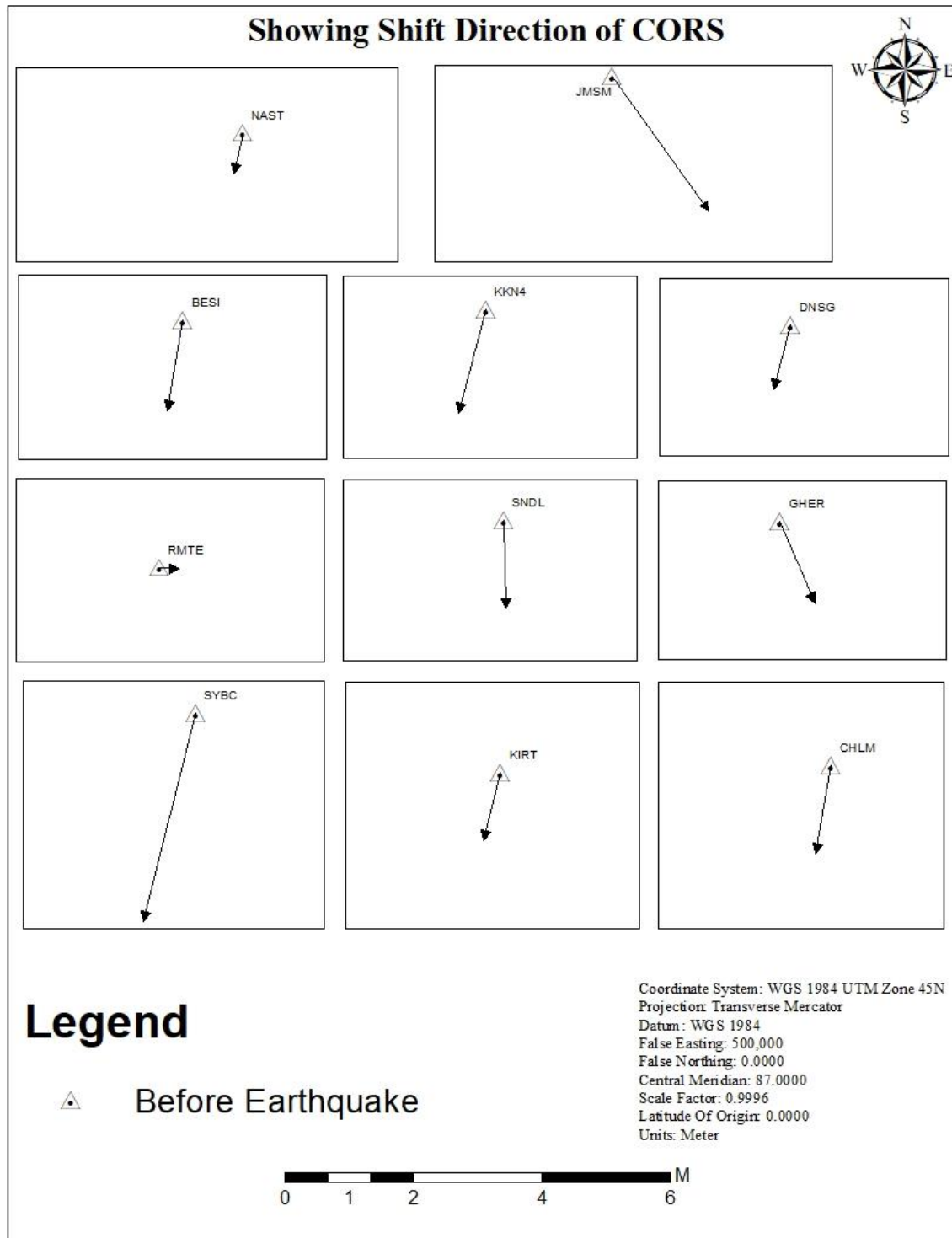


Figure 8: Showing Shift Direction Map of CORS

5. Discussion

The earthquake also had a significant impact on the CORS itself. Several of the stations were damaged or destroyed during the earthquake, which disrupted the accuracy and availability of positioning data in the affected areas. This disruption had a significant impact on a range of activities, including transportation, surveying, and disaster management.

In the aftermath of the earthquake, the Nepal government and international organizations worked to repair and restore the damaged CORS. This effort was critical in enabling the country to resume normal operations and rebuild infrastructure, as accurate positioning information was essential for activities such as road construction and land surveying.

The earthquake also highlighted the importance of investing in resilient infrastructure, including the CORS network, to better prepare for future disasters. The Nepal government has since taken steps to strengthen the network and improve its resilience, including the installation of new stations and the implementation of new technologies.

Similarly, The eastern part of the epicenter (Barpak) was hugely affected by the earthquake whereas the western part had minor effect and the station DNSG had the minimum vertical shift of -0.001m and the station NAST had the maximum vertical shift of -1.824m and the station SNDL had the minimum horizontal shift of 0.046m and the station KKN4 had the maximum horizontal shift of -0.469m because of the Gorkha Earthquake 2015 and the aftershock.

Unequal shifts in affected regions central part of the study area has shifted in the south east direction.

Overall, the impact of the 2015 earthquake on the CORS network in Nepal was significant, but the network played a critical role in enabling the country to respond effectively to the disaster. The experience also highlighted the need for continued investment in resilient infrastructure to better prepare for future disasters and support long-term development goals.

5. CONCLUSION AND RECOMMENDATION

The massive Gorkha Earthquake has caused the changes of the ground surface and also on the permanently established control points.

The 2015 earthquake of Nepal has caused more shift in KKN4 Station. However, there are unequal shifts in the affected regions due to unequal stress on Earth's surface. Results of the project shows that all the CORS Station are negatively shifted in vertical direction.

For the determination of more efficient, accurate and reliable shifts of the surface due to various seismic activities, following points are recommended:

- The use of fixed control points should be discouraged to determine the deformation of the surface because the coordinates of the point do not remain same for a long time. Updating large network of fixed control points is time consuming.
- A series of network of CORS should be established in Nepal more densely so that the interpolation of unknown locations could be estimated properly.
- The network adjustment can be done by using shortest possible unaffected base stations.
- The data used should be continuous over a large period of time for better result.

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7. Appendix

The website interface that was used to download data is

<https://www.unavco.org/data/gps-gnss/data-access-methods/dai1/dai>

The screenshot shows the GAGE (Global Access to Geospatial and Earthquake) website interface. The header includes the GAGE logo, EarthScope Consortium branding, and navigation links for DATA, INSTRUMENTATION, SOFTWARE, KNOWLEDGE BASE, WHAT WE DO, EVENTS, COMMUNITY, EDUCATION, and NEWS. The breadcrumb trail indicates the current location: home > data > gps/gnss > data access methods > dai1 > perm sta.php. The main title is "Data Archive Interface v1". On the left, there is a "Data" sidebar with links to "Help with Data", "GPS/GNSS Data", "Data Access", "Search and Access Methods", "File Server", "File Server Access Examples", "Web Services", "Real-time GPS/GNSS Data", "Derived Data Products", "Associated Data Products", "Data Formats", "Data Management", and "Data Operations". Below the sidebar is a "Related Links" section with a link to "Data Policy". The main content area features a "Permanent Stations" section with tabs for "Station Search", "Temporal Search", "Spatial Search", "Monument/Equipment Search", "Map Display", "Tabular Display", "Help", and "Minimize". The "Station Search" tab is active, showing a search interface with a "Filter by 4-char Station Code List" (containing 1LSU - Louisiana State University, 1NSU - Northwestern State University, 1ULM - U of Louisiana-Monroe, and 299C - 2999 Continuous), a "Filter by Network" section (with "Contains" and "Equals" radio buttons), and a "Filter By Operational Status" section (with checkboxes for Active, Inactive, Retired, Pending, and Intermittent 5Hz or 1Hz). A "Please provide feedback" and "Printable View" button are also visible.

home > data > gps/gnss > data access methods > dai1 > perm sta.php

Data

- Help with Data
- **GPS/GNSS Data**
 - Data Access
 - Search and Access Methods
 - File Server
 - File Server Access Examples
 - Web Services
 - Real-time GPS/GNSS Data
 - Derived Data Products
 - Associated Data Products
 - Data Formats
 - Data Management
 - Data Operations

Related Links

- Data Policy

Data Archive Interface v1

Please provide feedback Printable View

Permanent Stations

Station Search Temporal Search Spatial Search Monument/Equipment Search Map Display Tabular Display Help Minimize

☐ Filter by 4-char Station Code List:

Add selection(s) to the List Clear List

1LSU - Louisiana State University
1NSU - Northwestern State University
1ULM - U of Louisiana-Monroe
299C - 2999 Continuous

☐ Filter by Network:

Contains: ☒ Equals: ☐

Afar
Afar High Rate
Africa Array
Akutan

Filter By Operational Status:

- ☒ Active
- ☒ Inactive
- ☒ Retired
- ☒ Pending
- ☐ Intermittent 5Hz or 1Hz

3.4 Positional Uncertainty (95% C.L.) - Geodetic, ITRF2014

Station	Longitude(East) (m)	Latitude(North) (m)	Ellipsoidal Height(Up) (m)
BESI	0.007	0.005	0.013
CHLM	0.006	0.005	0.012
DNSG	0.006	0.005	0.015
GHER	0.008	0.005	0.018
JMSM	0.006	0.005	0.012
KIRT	0.007	0.005	0.015
KKN4	0.007	0.005	0.013
NAST	0.007	0.005	0.014
RMTE	0.006	0.005	0.011
SNDL	0.006	0.005	0.012
SYBC	0.007	0.005	0.013
CHUM	0.006	0.005	0.010
DSMG	0.010	0.005	0.016
FOMO	0.010	0.005	0.016
HYDE	0.006	0.005	0.011
IISC	0.007	0.005	0.013
KIT3	0.006	0.005	0.012
LCK3	0.006	0.005	0.012
LHAZ	0.006	0.005	0.011
MSHN	0.009	0.007	0.025
PBRI	0.007	0.006	0.013
POL2	0.006	0.005	0.010
TOKA	0.008	0.006	0.016
URUM	0.006	0.005	0.012

4 Ambiguity Resolution - Per Baseline

Baseline	Ambiguities Resolved	Baseline Length (km)
FOMO - RMTE	36.8 %	2753.203
RMTE - SNDL	86.3 %	90.364
CHLM - SNDL	83.3 %	102.945
RMTE - URUM	95.9 %	1861.757
CHUM - POL2	89.3 %	35.732
KIT3 - MSHN	78.5 %	722.387
CHLM - JMSM	87.0 %	167.534
CHLM - DNSG	84.6 %	152.961
LCK3 - RMTE	89.6 %	560.118
RMTE - SYBC	82.1 %	91.997
KKN4 - NAST	76.5 %	16.690
KIRT - NAST	91.2 %	4.800
CHUM - KIT3	76.5 %	787.202
HYDE - RMTE	78.8 %	1343.007
DSMG - FOMO	85.6 %	5.048
CHLM - KKN4	90.2 %	45.219
LHAZ - SNDL	89.8 %	577.112
CHUM - URUM	90.2 %	1042.674
PBRI - RMTE	81.3 %	1810.214
BESI - CHLM	91.7 %	91.817
PBRI - TOKA	80.0 %	1047.628
BESI - GHER	82.6 %	16.468
HYDE - IISC	68.2 %	497.626
AVERAGE	82.4%	601.065

Please note for a regional solution, such as used by AUSPOS, ambiguity resolution success rate of 50% or better for a baseline formed by a user site indicates a reliable solution.

5.2 Data Preprocessing and Measurement Modelling

Data preprocessing	Phase preprocessing is undertaken in a baseline by baseline mode using triple-difference. In most cases, cycle slips are fixed by the simultaneous analysis of different linear combinations of L1 and L2. If a cycle slip cannot be fixed reliably, bad data points are removed or new ambiguities are set up. A data screening step on the basis of weighted postfit residuals is also performed, and outliers are removed.
Basic observable	Carrier phase with an elevation angle cutoff of 7° and a sampling rate of 3 minutes. However, data cleaning is performed at a sampling rate of 30 seconds. Elevation dependent weighting is applied according to $1/\sin(e)^2$ where e is the satellite elevation.
Modelled observable	Double differences of the ionosphere-free linear combination.
Ground antenna phase centre calibrations	IGS14 absolute phase-centre variation model is applied.
Tropospheric Model	A priori model is the GMF mapped with the DRY-GMF.
Tropospheric Estimation	Zenith delay corrections are estimated relying on the WET-GMF mapping function in intervals of 2 hour. N-S and E-W horizontal delay parameters are solved for every 24 hours.
Tropospheric Mapping Function	GMF
Ionosphere	First-order effect eliminated by forming the ionosphere-free linear combination of L1 and L2. Second and third effect applied.
Tidal displacements	Solid earth tidal displacements are derived from the complete model from the IERS Conventions 2010, but ocean tide loading is not applied.
Atmospheric loading	Applied
Satellite centre of mass correction	IGS14 phase-centre variation model applied
Satellite phase centre calibration	IGS14 phase-centre variation model applied
Satellite trajectories	Best available IGS products.